417522 — DATA MODELING AND VISUALIZATION

UNIT II — TESTING AND DATA MODELING

Expanded In-Semester Examination Notes • Fourth Year AI & DS • SPPU 2020 Course

 SYLLABUS-LOCKED
Unit II in the official syllabus contains: Random Numbers and Simulation; Hypothesis Testing; and Stochastic Processes and Data Modeling, followed by the two listed hypothesis-testing case studies. filecitefturn25file7

SYLLABUS POINT	COVERED
Sampling of continuous distributions	✓
Monte Carlo methods	✓
Type I and Type II errors	✓
Rejection regions	✓
Z-test	✓
T-test	✓
F-test	✓
Chi-Square test	✓
Bayesian test	✓
Markov process	✓
Hidden Markov Models	✓
Poisson Process	✓
Gaussian Processes	✓
Auto-Regressive process	✓
Moving average process	✓
Bayesian Network	✓
Regression	✓
Queuing systems	✓
Dieters vs exercisers	✓
New medicine testing	✓

UNIT II — MASTER MEMORY MAP

 REMEMBER
Random sampling → Simulation → Hypothesis testing → Stochastic processes → Probabilistic data modeling → Regression/Queuing

PART A — RANDOM NUMBERS AND SIMULATION

1. RANDOM NUMBERS

A random number is generated so that the values have the required random behavior/probabilities. Random numbers are useful for representing uncertainty in simulations.

• Uniform random numbers are commonly used as a starting point.

• Random values can be transformed to obtain samples from other distributions.

• Repeated random sampling lets us study a system without physically repeating the real experiment.

2. SAMPLING FROM CONTINUOUS DISTRIBUTIONS

Sampling from a continuous distribution means generating observations that behave according to a chosen continuous probability model. A generated value can lie anywhere within the distribution's continuous range.

Monte Carlo

Basic idea

• Choose the required distribution.

• Generate random values according to that distribution.

• Treat the generated values as simulated observations.

• Analyze the simulated sample.

3. SIMULATION

Simulation means creating a model of a real system and carrying out repeated computational experiments to understand how the system behaves. It is useful when a direct analytical solution is difficult, expensive or impossible.

• Customer-service waiting lines.

• Traffic and network behavior.

• Financial uncertainty.

• Reliability and risk.

• Inventory/resource planning.

4. MONTE CARLO METHODS

Monte Carlo methods are computational algorithms based on repeated random sampling to obtain numerical results. The uploaded lecture notes emphasize their use for optimization, numerical integration and generating draws from probability distributions. filecite turn26file1 turn26file2

Monte Carlo

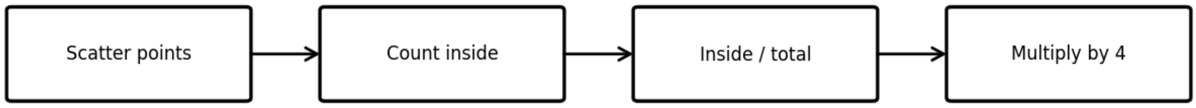
Typical pattern

- Define a domain or problem.
- Generate inputs randomly from an appropriate probability distribution.
- Perform the required computation for each random input.
- Aggregate the results to obtain an estimate.

MONTE CARLO — CLASSIC π EXAMPLE

A quadrant is inscribed inside a unit square. The area ratio is π/4. If points are scattered uniformly over the square, the fraction that falls inside the quadrant estimates π/4. Multiplying that fraction by 4 gives an estimate of π.

Estimating π



Why it works: the random points sample the square uniformly. With many points, the observed fraction approaches the area ratio.

Advantages

• Flexible and practical for complicated probabilistic problems.

• Can handle high-dimensional systems.

• Useful when exact calculation is difficult.

• Easy to repeat with different random inputs.

Limitations

• Estimate has random simulation error.

• More trials generally improve precision but require more computation.

• Results depend on the quality of the model and random sampling.

EXAM MEMORY
MONTE CARLO = RANDOM SAMPLES → REPEAT MANY TIMES → AGGREGATE/ESTIMATE

PART B — HYPOTHESIS TESTING

5. BASIC IDEA OF HYPOTHESIS TESTING

Hypothesis testing is a statistical method for using sample evidence to evaluate a claim about a population. The process begins with competing assumptions and ends with a decision and a conclusion in the language of the original problem. filecite turn26file14

6. NULL AND ALTERNATIVE HYPOTHESES

• H_0 (null hypothesis): default/status-quo claim being tested.

• H_1 or H_a (alternative hypothesis): competing claim that the evidence may support.

Example — new medicine: H_0 = no improvement over comparison treatment; H_1 = medicine improves the outcome.

7. SIGNIFICANCE LEVEL α

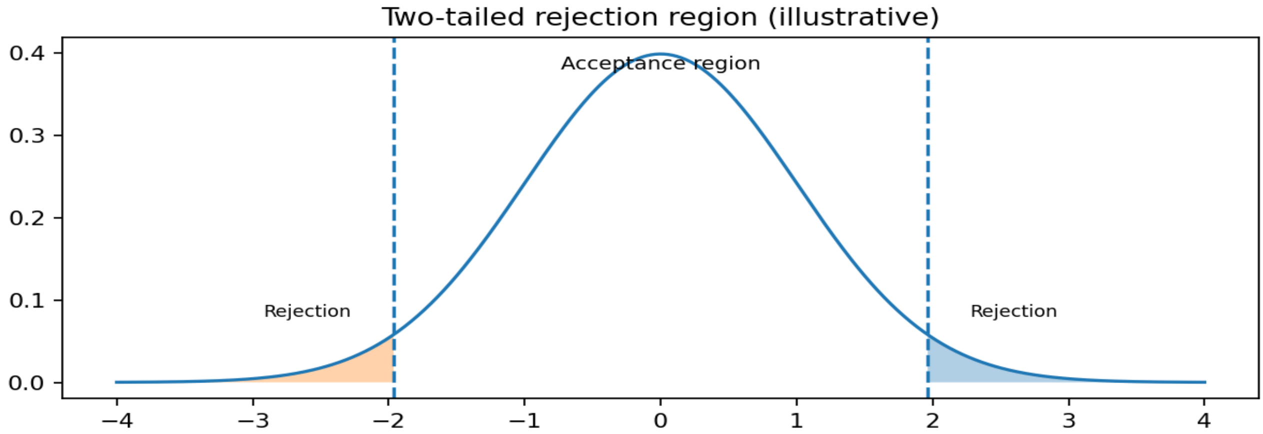
α is the chosen threshold for deciding that evidence is sufficiently extreme to reject H_0. A common classroom value is 0.05.

8. p-VALUE

The p-value measures how compatible the observed result, or a more extreme result, is with H_0. If p-value < α, reject H_0. If p-value ≥ α, do not reject H_0.

9. REJECTION REGION

The rejection region is the part of the test-statistic distribution where the statistic is sufficiently extreme to lead to rejection of H_0. The position of this region depends on α and on whether the alternative is one-tailed or two-tailed.



Alternative	Typical rejection location
Right-tailed	Extreme upper/right side
Left-tailed	Extreme lower/left side
Two-tailed	Both extreme tails

10. TYPE I AND TYPE II ERRORS

Reality / Decision	Reject H_0	Do not reject H_0
H_0 true	Type I error (false positive)	Correct decision
H_0 false	Correct decision	Type II error (false negative)

Type I error probability is α. Type II error probability is β.

MEMORY
TYPE I = reject a TRUE H_0 → false positive. TYPE II = fail to reject a FALSE H_0 → false negative.

11. GENERAL STEPS OF A HYPOTHESIS TEST

- State the problem.
- Write H_0 and H_1.
- Check relevant assumptions and sample information.
- Select the appropriate test.
- Choose significance level α.
- Calculate the test statistic.
- Find p-value or compare with critical/rejection region.
- Reject or do not reject H_0.
- Write the conclusion in the original problem context.

EXAM POINT
Do not end only with “reject H_0.” Always add a sentence explaining what that means for the actual problem.

12. Z-TEST

A Z-test uses the standard normal distribution for an appropriate hypothesis test. In the common one-sample mean setting, the population standard deviation σ is known.

$$\mathbf{z=(\bar{X}-\mu_0)/(\sigma/\sqrt{n})}$$

• Used when the normal approximation/assumptions are appropriate.

• σ is known in the standard one-sample z formulation.

• The alternative determines the rejection direction.

13. T-TEST

A t-test is used for mean-based hypothesis testing when the population standard deviation is unknown and is estimated from the sample. The t distribution accounts for the extra uncertainty from estimating σ using s.

$t=(\bar{X}-\mu_0)/(s/\sqrt{n})$, $df=n-1$ for a one-sample t-test

• One-sample t-test: compare a sample mean with a hypothesized mean.

• Two-sample t-test: compare means of two groups under relevant assumptions.

• As degrees of freedom increase, the t distribution approaches the normal distribution.

14. Z-TEST vs T-TEST

Point	Z-test	T-test
Distribution	Standard normal	t distribution
Population σ	Known in standard formulation	Unknown; use sample s
Common use	Mean testing under known σ	Mean testing with estimated σ
Statistic	$(\bar{X}-\mu_0)/(\sigma/\sqrt{n})$	$(\bar{X}-\mu_0)/(s/\sqrt{n})$
Memory	Z → known σ	T → use s

VERY IMPORTANT
Do not choose a Z-test merely because n is large. The variance/assumption context matters.
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15. F-TEST

The F-test is based on the F distribution and is commonly used for comparing variances and in variance-ratio settings.

Typical two-variance statistic: $F=s_1^2/s_2^2$

• F values are non-negative.

• The distribution uses two degrees of freedom.

• The rejection region depends on the alternative hypothesis.

• For exam purposes, remember: F-test → variance/variance ratio.

16. CHI-SQUARE TEST

The Chi-square test compares observed and expected frequencies under a null hypothesis.

$$\chi^2=\Sigma(O-E)^2/E$$

• O = observed frequency.

• E = expected frequency under H_0.

• Goodness-of-fit: checks whether observed categorical frequencies fit a specified distribution/proportion.

• Test of independence: checks whether two categorical variables are associated.

17. BAYESIAN TESTING

Bayesian testing combines prior information with observed data to obtain a posterior assessment of a hypothesis or parameter.

Bayes theorem: P(H|D)=[P(D|H)P(H)]/P(D)

Term	Meaning
Prior P(H)	Belief before current data
Likelihood P(D|H)	How compatible the data are with the hypothesis
Evidence P(D)	Overall probability of observed data
Posterior P(H|D)	Updated belief after data

MEMORY
PRIOR + DATA → POSTERIOR

PART C — STOCHASTIC PROCESSES AND DATA MODELING

18. STOCHASTIC PROCESS — BASIC IDEA

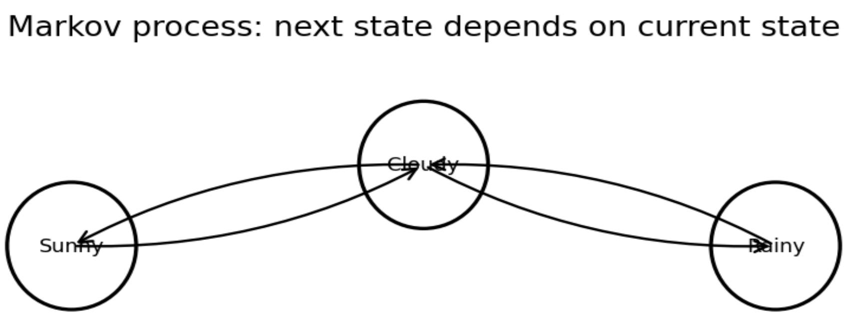
A stochastic process is a collection of random variables indexed by time or another index. It represents a system whose behavior changes with uncertainty.

Examples: weather state, queue length, sensor readings, demand over time and random movement.

19. MARKOV PROCESS

A Markov process is a stochastic process satisfying the Markov property: once the present state is known, the future does not require the complete past history for its transition probability.

$$P(X_{n+1}=x \mid X_n=x_n,\ldots,X_0=x_0)=P(X_{n+1}=x \mid X_n=x_n)$$



Example: Weather states may be Sunny, Cloudy and Rainy. A simplified Markov model assumes tomorrow's state is determined probabilistically from today's state rather than requiring every previous day's weather.

Important terms

• State: possible condition of the process.

• Transition probability: probability of moving from one state to another.

• Transition matrix: table of transition probabilities.

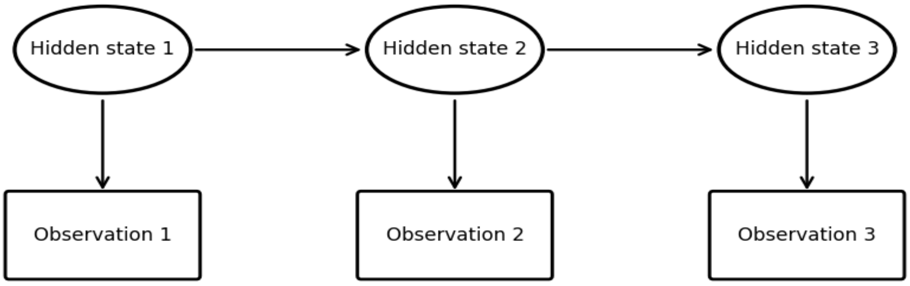
• Markov property: current state summarizes the information needed for the next transition.

MEMORY
MARKOV = FUTURE DEPENDS ON CURRENT STATE, NOT THE FULL PAST.

20. HIDDEN MARKOV MODEL (HMM)

An HMM is a stochastic model in which the underlying states are hidden, while observations generated by those states are visible.

Hidden Markov Model — hidden states generate observations



Component	Meaning
Hidden states	Underlying conditions that cannot be observed directly
Observations	Visible outputs generated by hidden states
Transition probabilities	Probability of moving between hidden states
Emission probabilities	Probability of an observation given a hidden state

Example: A wearable sensor records movement readings. The actual activity—walking, running or sitting—can be treated as hidden, while sensor measurements are observations. The HMM can infer likely hidden states.

□ EXAM
HMM = HIDDEN STATES + OBSERVATIONS + TRANSITIONS + EMISSIONS

21. POISSON PROCESS

A Poisson process models random event occurrences over time under the standard Poisson-process assumptions.

Poisson process — event arrivals over time

Random event times; N(t) counts events by time t

For N(t), the number of events by time t: $P(N(t)=n)=e^{(-\lambda t)}(\lambda t)^n/n!$

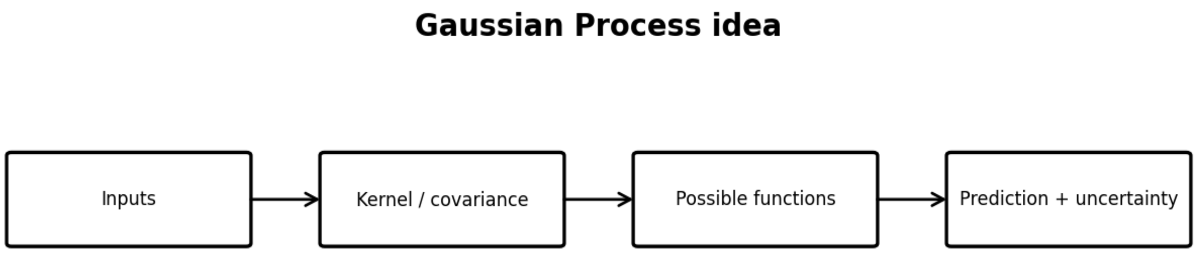
- λ = average event rate per unit time.
- Events in disjoint intervals are independent under the standard assumptions.
- Interarrival time is exponentially distributed in the standard Poisson-process model.

Example: number of customer calls received per hour or arrivals at a service counter.

⚠ **DO NOT CONFUSE**
Poisson distribution gives the count for an interval. Poisson process describes the event-arrival mechanism over time.

22. GAUSSIAN PROCESSES

A Gaussian Process (GP) is a non-parametric and probabilistic approach to data modeling. Instead of producing only one deterministic function, it places a probability distribution over possible functions/outputs.



- Probabilistic: gives prediction and uncertainty.
- Non-parametric: model complexity adapts to observed data.
- Kernel/covariance function: represents similarity/correlation between input points.
- Can be used for regression and classification.
- Useful for interpolation, Bayesian optimization, spatial data, time-series modeling and anomaly detection.

MEMORY
GAUSSIAN PROCESS = FUNCTION PREDICTION + UNCERTAINTY + KERNEL

23. AUTO-REGRESSIVE (AR) PROCESS

An Auto-Regressive process models the current time-series value using a linear combination of previous values of the same series.

$$\textbf{AR(1): } X_t = c + \varphi_1 X_{t-1} + \varepsilon_t$$
$$\textbf{AR(p): } X_t = c + \varphi_1 X_{t-1} + \ldots + \varphi_p X_{t-p} + \varepsilon_t$$

• p = number of previous observations used.

• φ values are model coefficients.

• ε_t is the random innovation/error term.

• Example: today's demand depends on yesterday's and earlier demand values.

24. MOVING AVERAGE (MA) PROCESS

An MA process models the current time-series value using the current and past random error terms.

$$\textbf{MA(q): } X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta q \varepsilon_{t-q}$$

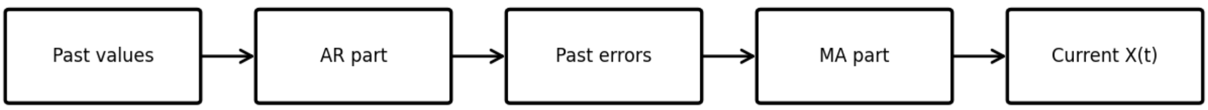
• q = number of past error terms included.

• μ = mean level in the common formulation.

• ε terms represent random shocks/errors.

• MA in this stochastic-process context does NOT simply mean a rolling average of raw observations.

Time-series dependency



25. AR vs MA

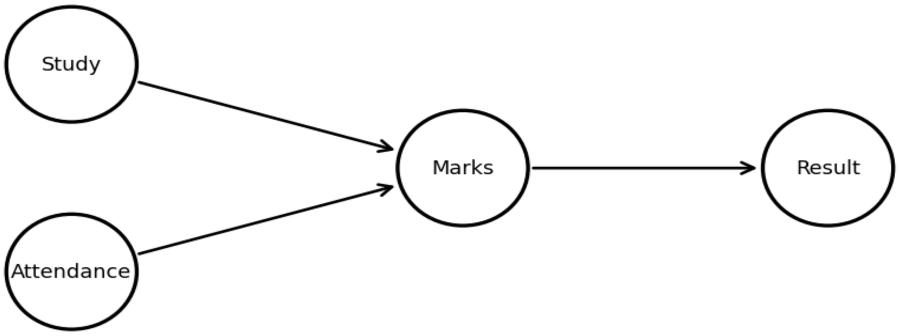
Point	AR	MA
Depends on	Past values of the series	Current/past error terms
Notation	AR(p)	MA(q)
Main idea	Past observations influence current value	Past shocks/errors influence current value
Memory	PAST VALUES	PAST ERRORS

□ VERY IMPORTANT
AR = past observations. MA = past error terms/shocks. □filecite□turn26file16□

26. BAYESIAN NETWORK

A Bayesian Network is a probabilistic graphical model where variables are represented by nodes and directed dependencies by arrows. The graph is a Directed Acyclic Graph (DAG).

Bayesian Network — directed acyclic dependencies



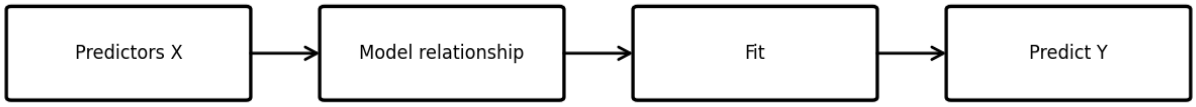
Element	Meaning
Node	Variable
Directed edge	Dependency relationship
DAG	Directed graph with no directed cycle
Conditional probability	Quantifies a node's relationship with parent nodes

Example: Study and Attendance influence Marks; Marks influences Result. The network represents the dependency structure and supports probabilistic reasoning.

27. REGRESSION

Regression models the relationship between a dependent/target variable and one or more independent/predictor variables. In the standard regression setting, the target is quantitative/continuous.

Regression



Simple linear regression

$$\mathbf{Y=\beta_0+\beta_1 X}$$

- Y = dependent/target variable.
- X = independent/predictor variable.
- β_0 = intercept.
- β_1 = slope.

Example: predict marks from study hours. If β_1 is positive, the fitted model predicts higher marks as study hours increase, subject to the model and data.

Multiple linear regression

$Y=\beta_0+\beta_1X_1+\beta_2X_2+\ldots+\beta pXp$

Example: predict house price using area, number of rooms and age.

Least-squares idea

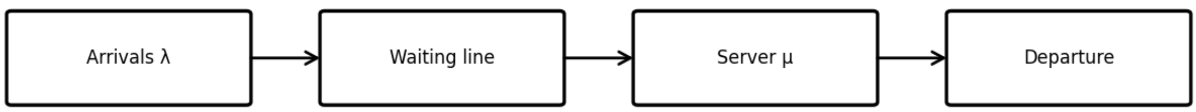
A common fitting method chooses coefficients that minimize the sum of squared residuals: $SSE=\Sigma(\text{actual}-\text{predicted})^2$.

> **□ MEMORY**
> **REGRESSION = USE X VARIABLES TO ESTIMATE/PREDICT Y**

28. QUEUING SYSTEMS

A queuing system represents entities that arrive, wait when necessary, receive service and then depart. It is useful for studying congestion, waiting time and service capacity.

Queuing system



Main components

- Arrival process: how customers/jobs enter.

- Queue: waiting line.
- Server/service facility: performs service.
- Service process: describes service time/rate.
- Queue discipline: rule for selecting the next customer, e.g., first-come-first-served.
- Departure: completed customer/job leaves the system.

M/M/1 queue

M/M/1 is a standard queueing model with Poisson-type arrivals, exponential service times and one server.

LITTLE'S LAW

Little's Law connects the average number of items in a system, average arrival rate and average time spent in the system.

$$L=\lambda W$$

Symbol	Meaning
L	Average number of entities in the system
λ	Average arrival rate
W	Average time an entity spends in the system

Example: If customers arrive at an average rate of 10 per hour and spend 0.5 hour in the system, then $L=10\times0.5=5$ customers on average.

For the standard M/M/1 stability condition: $\rho=\lambda/\mu$ and $\lambda<\mu$ (equivalently $\rho<1$).

> **VERY IMPORTANT**
> **Previous-year style question: “Explain the queuing system and illustrate Little's Law with a neat graph.” Draw Arrivals → Queue → Server → Departure and write $L=\lambda W$.**

29. CASE STUDY — DIETERS LOSE MORE FAT THAN EXERCISERS

The purpose is to use sample evidence to evaluate the claim that dieters lose more fat than exercisers.

• Define the two groups.

• H_0: average fat loss for dieters is not greater than for exercisers.

• H_1: average fat loss for dieters is greater than for exercisers.

• Collect representative sample data.

• Choose an appropriate two-group mean test based on the assumptions/design.

• Calculate the test statistic and p-value.

• If p-value < α, reject H_0 and conclude that the sample provides statistical evidence supporting greater average fat loss for dieters.

> **REMEMBER**
> **Statistical significance does not automatically mean the practical difference is large or important.**

30. CASE STUDY — NEW MEDICINE TESTING

A new medicine can be compared with a placebo or an existing treatment using sample data and a suitable hypothesis test.

• H_0: no meaningful difference in the population outcome.

• H_1: the new medicine changes/improves the outcome.

• Collect representative observations.

• Select a suitable test based on response and study design.

• Calculate statistic and p-value.

• Make the decision and explain it in the context of the medicine.

Error examples: Type I → conclude the medicine works when it actually does not. Type II → fail to detect a real benefit.

UNIT II — EXAM PRIORITY ▯

Priority	Question
▯ Very High	Monte Carlo method + π example + steps
▯ Very High	Hypothesis testing steps + H_0/H_1 + α + p-value + rejection region
▯ Very High	Type I and Type II errors
▯ Very High	Z-test vs T-test
▯ Very High	Poisson process with example
▯ Very High	HMM with diagram and components
▯ Very High	AR and MA processes / difference
▯ Very High	Bayesian Network with example + diagram
▯ High	F-test
▯ High	Chi-square test
▯ High	Bayesian test
▯ High	Markov process
▯ High	Gaussian Process
▯ High	Regression
▯ High	Queuing system + Little's Law
▯ High	Both syllabus case studies

UNIT II — DO NOT CONFUSE

Pair	Correct distinction
Type I vs Type II	Type I = reject true H_0; Type II = fail to reject false H_0
Z vs T	Z commonly uses known σ; T estimates σ using sample s
F vs Chi-square	F → variance ratio; Chi-square → observed/expected frequencies and independence
Markov vs HMM	Markov states are modeled directly; HMM states are hidden and generate observations
Poisson distribution vs process	Distribution = count in interval; process = event occurrence over time
AR vs MA	AR → past values; MA → past error terms
Bayesian Network vs HMM	BN = general directed probabilistic dependencies; HMM = sequential hidden-state model
Regression vs classification	Regression predicts quantitative target; classification predicts categories

UNIT II — FINAL MEMORY CHART ▯

Topic	One-line memory
Random numbers	Represent uncertainty
Continuous sampling	Generate observations from a continuous distribution
Monte Carlo	Random samples → repeat → estimate
H_0/H_1	Default claim vs competing claim
α	Significance level
p-value	Compatibility/evidence measure against H_0
Rejection region	Extreme statistic values leading to rejection
Type I	False positive
Type II	False negative

Topic	One-line memory
Z	Known σ / normal setting
T	Unknown σ → use s
F	Variance ratio
Chi-square	Observed vs expected frequencies
Bayesian	Prior + data → posterior
Markov	Future depends on current state
HMM	Hidden states + observations
Poisson process	Events over time
Gaussian Process	Function prediction + uncertainty + kernel
AR	Past values
MA	Past errors
Bayesian Network	Directed probabilistic dependencies
Regression	Estimate/predict Y from X
Queue	Arrival → wait → service → departure
Little's Law	$L=\lambda W$

✍ 5-MARK ANSWER FORMULA
Definition → explanation of components → formula/steps → example → pictorial diagram where appropriate → final key point.

□ SYLLABUS CONTROL
The official Unit-II syllabus explicitly lists all topics covered above. The uploaded lecture notes provide the detailed framing for Monte Carlo, hypothesis testing, Markov property, GP, AR/MA, HMM, Bayesian Network, regression and queuing.
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